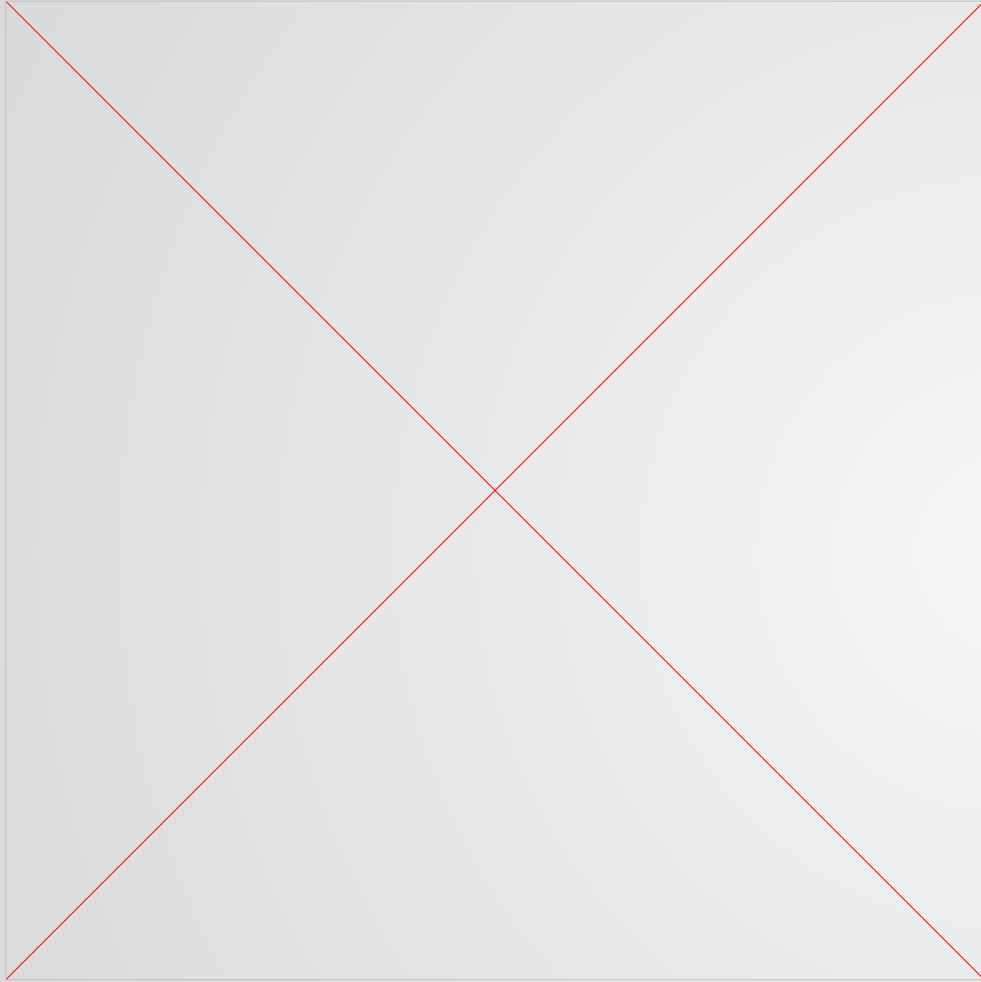




Machine Learning

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What is Machine Learning

- Certain tasks are extremely difficult to program by hand:



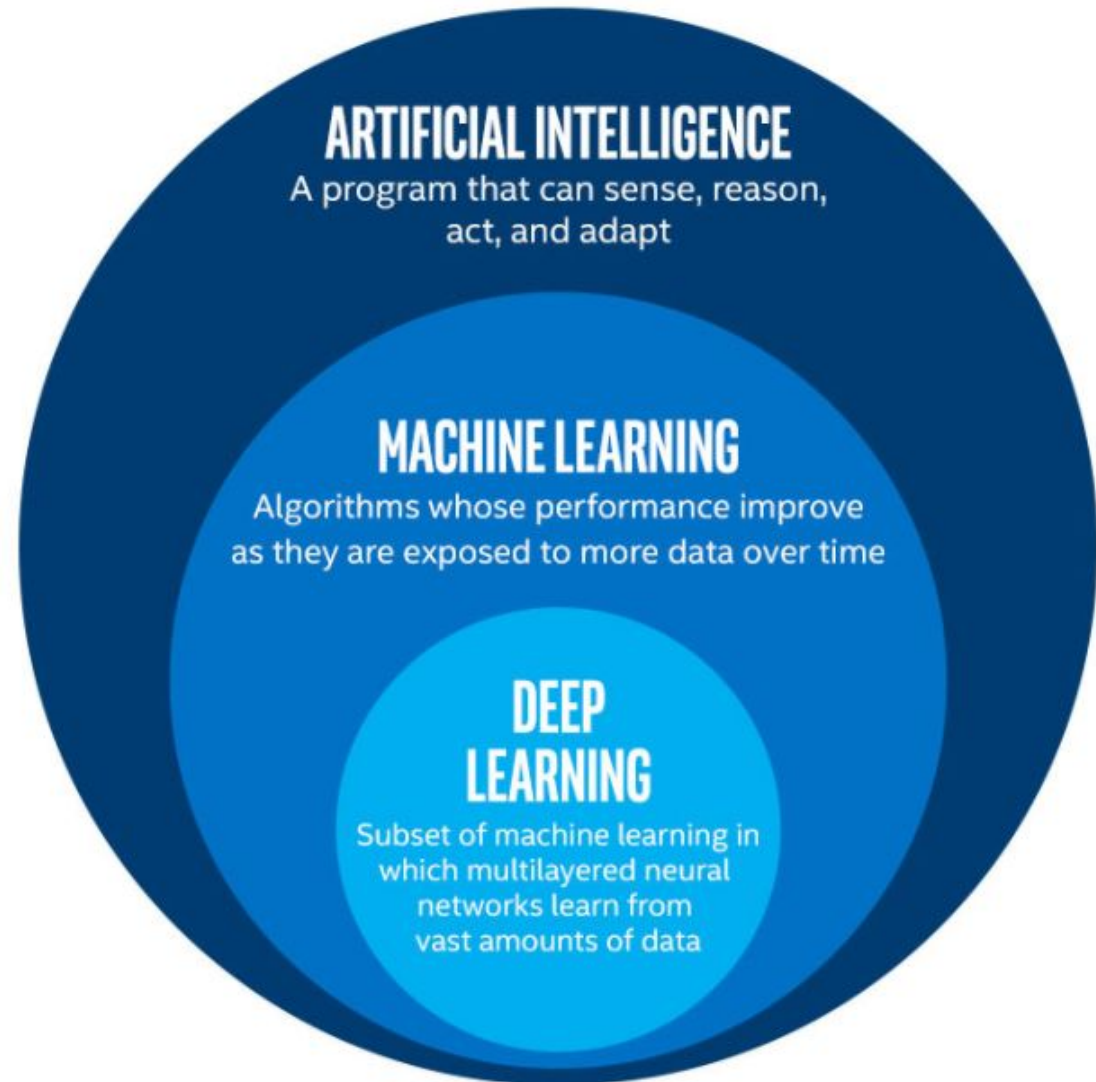
Spam filtering

What is Machine Learning

Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed.

It automates analytical modeling, by using algorithms that iteratively learn from the data without being explicitly programmed.

Machine Learning



Machine Learning

- Virtually all learning problems can be formulated as (complex) mappings between inputs and outputs
- We are trying to learn what is the best output $\mathbf{o}$ to produce for each possible input $\mathbf{i}$
- Mathematically speaking, we search for a «good» function $\mathbf{F}: \mathbf{I} \rightarrow \mathbf{O}$, where $\mathbf{I}$ is the set of possible inputs, and $\mathbf{O}$ the set of possible outputs

Machine Learning

	Input i	Output o
Spam filtering	An email	{spam, non-spam}
Face recognition	An image	Identified faces
Machine translation	A sentence in language A	A sentence in language B
Speech recognition	A speech signal	A (text) sentence
Data mining	A financial transaction	{fraud, non-fraud}
Robot motion	Sensory data	Motor control

Learning Methods

- But how do we learn this mapping?
- The learning method depends on the kind of data that we have at our disposal
 - We can have examples of data where we have both the inputs and outputs: $(\mathbf{i}, \mathbf{o})$
 - For some data, we only have the inputs $\mathbf{i}$

Learning Methods

- But how do we learn this mapping?
- The learning method depends on the kind of data that we have at our disposal
 - We can have examples of data where we have both the inputs and outputs: (i, o)
 - supervised learning
 - For some data, we only have the inputs i
 - unsupervised learning

Types of Machine Learning

Supervised

- Regression
- Classification

Unsupervised

- Clustering
- Dimensionality Reduction

Semi-Supervised

- Active Learning

Case-Based Reasoning

Reinforcement Learning

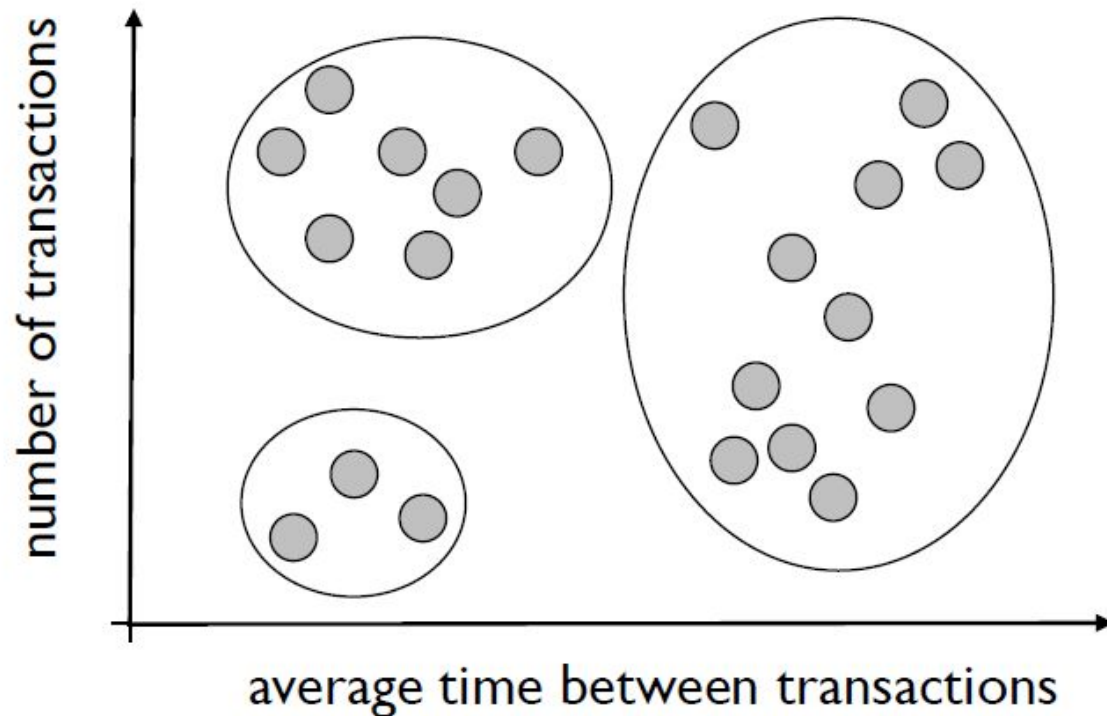
Supervised Learning

- In supervised learning, we have *training data* encoded as pairs (i,o) , where the «correct» output is often manually annotated
 - E.g. spam filtering, machine translation, face recognition, etc.
- The function $\mathbf{F}: \mathbf{I} \rightarrow \mathbf{O}$ is often dependent on a (sometimes large) set of parameters
- ... and the learning goal is to «adjust» these parameters in order to fit the data

Supervised Learning

- A good learning model is a model that **generalises** well to new data
 - In other words, it is able to *abstract* over its experience to detect underlying patterns
 - The design and test of such models is a crucial part of machine learning
- Else, the model is said to be *overfitted*
 - In other words, it is very well «fitted» to the examples it has processed, but perform very poorly with unseen data

Unsupervised Learning



- Sometimes, we don't have access to any output value $\mathbf{o}$, we simply have a collection of input examples $\mathbf{i}$
- In this case, what we try to do is to learn the *underlying patterns* of our data
 - is there any *correlations* between features?
 - can we *cluster* our data set in a few groups which behave similarly, and detect *outliers*?

Algorithms

Supervised Learning

Linear Regression

Logistic Regression

Support Vector Machines

K Nearest Neighbors

Decision Tree

Unsupervised Learning

K Means Clustering

Hierarchical Clustering

DBSCAN

Principal Component
Analysis

Reinforcement Learning

Q-Learning

SARSA

Monte Carlo

Deep Q Network

Applications

Supervised Learning



Weather Prediction



Sales Forecast



Stock Price Analysis

Unsupervised Learning



Customer Segmentation



Churn Analysis

Reinforcement Learning



Building Games



Training Robots